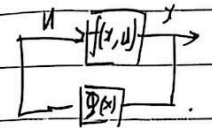
Advanced control (NLTI)

Advanced control — noliner part

$\dot{x} = f(x,u)$, if $u = \Phi(x)$



eg1: $\dot{x} = f(x,u) = ax^2 + u$, $u = \Phi(x)$

$f(0,0) = 0$, origin is fixed point

hope the ~~st~~ stability of origin

one choice: $u = -ax^2 - x$ → provide stability

↓ elimate the ~~N~~ noliner part

now: $\dot{x} = -x$, $x(t) = Ce^{-t}$

① a must be known

② so simple that ~~mab~~ maybe no the most good control

eg2: $\dot{x} = x^2 - x^3 + u$, $u_1 = -x^2 + x^3 - x$

an alternative method: Lyapunov Direct Method

the sum of thert ~~A~~: ① $\dot{x} = f(x)$, $\dot{x} = 0$, $x = 0$

② V: $V(0) = 0$, $V(x)$: PD $\dot{V}(x)$: ND

⇒ $x = 0$ is Asympotically stable

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now, Let $V = \frac{1}{2}x^2$, $V(x)$ PD

取 $V(0)=0$,

$\dot{V} = \frac{\partial V}{\partial x} \cdot \frac{\partial x}{\partial t} = x \cdot \dot{x} = x^3 - x^4 + xu$

Let $u_2 = -x^2 - x \Rightarrow \dot{V} = -x^4 - x^2$ ND

$u_3 = -x^2 \Rightarrow \dot{V} = -x^4$

u_1, u_2, u_3 all can make the system stable

反馈法:

Feedback linerization:

eg. $m\ddot{x} + \alpha x^3 = F$, object: change F (或 u), make the ball crawl along the ~~part~~ desired track

令: $F = u$, $x_1 = x$, $x_2 = \dot{x}$ $\therefore x_1 \to x_{1d}$

$\dot{x}_1 = \dot{x} = x_2$

$\dot{x}_2 = \ddot{x} = -\frac{\alpha}{m}x_1^3 + \frac{1}{m}u$

引入 e: error, $e = x_{1d} - x_1$, object: $e \to 0$.

One Choice: Find Lyapunov $V(e)$

~~eg.~~ $V_1 = \frac{1}{2}e^2$, PD, $V(0)=0$

$\dot{V}_1 = e\dot{e} = e(\dot{x}_{1d} - x_2) = -k_1 e^2$,

we hope $\dot{V}_1$ ND, so Let $\dot{x}_{1d} - x_2 = -k_1 e$

the $x_{2d} = \dot{x}_{1d} + k_1 e$

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new object: $x_2 \to x_{2d}$

$\delta = x_{2d} - x_2$, the $\dot{V}_1 = -k_1 e^2 + e\delta$

$\dot{\delta} = \ddot{x}_{1d} + k_1(\dot{x}_{1d} - x_2) + \frac{\alpha}{m}x_1^3 - \frac{1}{m}u$

object (new): $\delta, e \to 0 \Rightarrow V(e,\delta)$ PD $\dot{V}(e,\delta)$ ND

$V_2 = V_1 + \frac{1}{2}\delta^2$: PD

$\dot{V}_2 = \dot{V}_1 + \delta\dot{\delta} = -k_1 e^2 + \delta(e + \dot{\delta})$

resemble the last part of $\dot{V}_1$

$e + \dot{\delta} = -k_2\delta$

then $u = me + m\ddot{x}_{1d} + mk_1(\dot{x}_{1d} - x_2) + \alpha x_1^3 + mk_2\delta$

the core of this u is avoid the emergence of dot e, dot delta, dot x_{2d} this unknow valve

now: $\dot{e} = -k_1 e + \delta$, $\dot{\delta} = -e - k_2\delta$

Feedback linearization

反步法的原则:

找出互相影响的关系，如上文为 $u \to x_2 \to x_1$

故 step1: find x_{2d} 使得 $x_1 \to x_{1d}$

step2: find u 使 $x_2 \to x_{2d}$

则 step1 的原则 $x_1 \to x_{1d}$ 则 $e = x_{1d} - x_1$, $e \to 0$

设计 $V(e)$ 即可则可得与 x_2 有关的表达，此处 x_2 为 x_{2d}

由 x_{2d} 到 u 同理. tip: $V(e, \delta)$ 关于 e, δ, 这样才能 stable

Nolinear Adaptive Controller

what happen if $(\dot{x}_1 = ax^2 + u)$ 中 a unknown

How to deal with it

Assuption: a is ~~contra~~ constant ~~and~~ just $\dot{a} = 0$

$\dot{e} = \dot{x}_d - ax^2 - u$

estimate $\hat{a}$

estimate error $\tilde{a} = a - \hat{a}$ the $\dot{\tilde{a}} = -\dot{\hat{a}}$

$V(e, \tilde{a}) = \frac{1}{2}e^2 + \frac{1}{2}\tilde{a}^2$

$\dot{V} = e\dot{e} - \tilde{a}\dot{\hat{a}}$

$u = \dot{x}_d - \hat{a}x^2 + ke$

then: $\dot{V} = -ke^2 - \tilde{a}(ex^2 + \dot{\hat{a}})$

if $\dot{\hat{a}} = -ex^2 \Rightarrow V(e, \tilde{a})$ NSD

引理: ① $V \geq 0$ ② $\dot{V} \leq -g(t)$, $g(t) \geq 0$

③ $\dot{g}(t)$ 有界即 $g(t)$ 一致连续

则 $\lim_{t \to \infty} g(t) = 0$

则 $\dot{V} = -ke^2 \leq -ke^2$ $g(t) = ke^2$ PSD

$\dot{g}(t) = 2kee \Rightarrow \lim_{t \to \infty} ke^2 = 0 \Rightarrow e \to 0 |_{t \to \infty}$

则 object 满足即 $e \to 0$

故 $\hat{a} = -\int_0^t ex^2 dt$

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Robust Control (sliding mode)

To achieve robust performance or stability in presence of bounded modeling error

in Adaptive control: a is constant

in Robust control: $|a-\bar{a}|\le\delta$ ~~$|a|\le|\bar{a}|$~~ / $|a|\le|\bar{a}|$

General: $\dot{x}=f(x)+u$, $x\to x_d$; $|f(x)|<\rho(x)$

$\dot{e}=\dot{x}_d-f(x)-u$

Let $u=\dot{x}_d+ke+\rho\,\mathrm{sgn}(e)$

proof of the stability

$V(e)=\frac{1}{2}e^2$, $\dot{V}(e)=-ke^2-ef(x)-\rho|e|$

$\le -k\cdot e^2+|e||f(x)|-\rho|e|$

$\le -ke^2=-2k\cdot V(e)$

引入 $S(t)>0$, $\dot{V}(e)+2kV(e)=-S(t)$

则 $V(t)=V(0)\exp(-2kt)-\exp(-2kt)\int_0^t \exp(2kt)S(t)dt$

$\le V(0)\exp(-2kt)$

则 $|e(t)|\le|e(0)|\exp(-kt)$ 指数渐近稳定

则 $e\to 0|_{t\to\infty}$

现在: $\dot{e}=-ke-f(x)-\rho(x)\frac{e}{|e|}$

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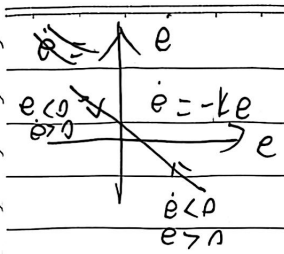

$-f(x)-\rho(x)\frac{e}{|e|}$ 控制项

$|f(x)|=|a|x^2\le|\bar{a}|x^2<\underbrace{|\bar{a}|(x^2+0.1)}_{\rho(x)}$

SMC的核心为找到一个滑模面使得设计u(t)使得系统稳定, SMC部分则是对于滑模面的寻找。

由结论u的频带特征该方法成立

$\dot{e}$的最终表达表明了在不同的情况下 $\rho(x)$会将$f(x)$带来的负作用消除

不过沿$\dot{e}=-ke$仅限于$f(x)>0$时的情况,

但$f(x)<0$且$e<0$时实际为向$\dot{e}=-ke\pm c$之间交换的滑动,依旧保持了系统误差在一定范围内

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High Gain and High Frequency Controller

$u = ke + \dot{x} + \rho\frac{e}{|e|} \Rightarrow \dot{e} = -ke - f(x) - \rho\frac{e}{|e|}$

problem: 可能: step

对于controller 很难

记 $u_{aux} = \rho\frac{e}{|e|}$

auxiliary 辅助

$u_{aux2} = \frac{1}{\varepsilon}\rho^2 e$: High Gain

$u_{aux3} = \frac{\rho^2 e}{\rho|e|+\varepsilon}$: High Frequency

$\varepsilon > 0$

u_{aux2}: 用足够大的输入抵消不稳定性

$\left|\frac{\rho^2 e}{\rho|e|+\varepsilon}\right| < \rho$, $\varepsilon \neq 0$

u_{aux2}:

Let: $V = \frac{1}{2}e^2$

$\dot{V} = e\dot{e} = -ef(x) - ke^2 - \frac{1}{\varepsilon}\rho^2 e^2$

$\leq \rho|e| - ke^2 - \frac{1}{\varepsilon}\rho|e|^2$

$= -ke^2 + \rho|e|(1 - \frac{1}{\varepsilon}\rho|e|)$

case1: $\rho|e| \geq \varepsilon \Rightarrow \dot{V} \leq -ke^2 \Rightarrow e \to 0|_{t\to\infty}$

case2: $\rho|e| < \varepsilon \Rightarrow \rho|e| \geq \rho|e|(1 - \frac{1}{\varepsilon}\rho|e|)$

$\Rightarrow \dot{V} \leq -ke^2 + \varepsilon = -2kV + \varepsilon$

Introduce $\delta(t) > 0$, $\dot{V} + 2kV + \varepsilon - \delta(t)$

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可得 $V(t) \leq V_0\exp(-2kt) + \frac{\varepsilon}{2k}(1 - \exp(-2kt))$

$\Rightarrow |e(t)| \leq \sqrt{\frac{\varepsilon}{k}}|_{t\to\infty}$

$\varepsilon \downarrow \to 0$, $e \to 0$ but $u \uparrow$

Globally uniformly ultimately bounded (GUUB)

u_{aux3} 本身作用

eg:

$\dot{x} = ax^2 + u$, $|a| \leq 1$, 参数不确定, $x \to x_d$, $\dot{x}_d = 0$

$f(x) = ax^2 \leq |a||x|^2 \leq x^2 < x^2 + 0.1 = \rho(x)$